

Detailed Router

- Input:
 - A DEF file containing the placement boundary, nets, boundary pins, and cell instances' locations
 - A GUIDE file containing the global routing solution
 - A LEF file containing the layer and macro information
 - Use `LEFDEFParser` to read the LEF/DEF files
 - Write a GUIDE file parser
- Output: DEF file containing the routes for each net
- Your submission should be a single python file `ROLLNo.py` containing a `detailed_route` routine with the arguments: `detailed_route(input DEF, input LEF, input GUIDE, output DEF)`.

DEF Parser

- DEFReader parses the .def file and stores them in the following data structures.

Class	Member	Type	Methods	Purpose
Point	x y	int int		– x coordinate – y coordinate
Rect	ll ur	Point Point		– lower-left coordinate – upper-right coordinate
Pin		str Point str	name origin print	– name of the pin – location of the pin – string containing the pin info
Net		str (str, str) str (str, int, int, int, int)	name pins print addRect	– Name of the net – list of pair of strings; (cell name, pin name in the cell); for boundary pins cell name string PIN – string containing the pin info – add a solution shape to the net; e.g. addRect("metal1", 100, 200, 300, 400)

DEF Parser cont..

Class	Type	Methods	Purpose
Grid	str int int int	orient x num step	– orientation 'X' or 'Y' – starting coordinate – number of grids + 1 – pitch of the grid
Component	str Point str str	name location orient macro setLocation(int,int) setOrient(str)	– instance name of the cell – lower left coordinate – orientation of the cell (N, S, FN, FS, ...) – type of the cell (defined in LEF file e.g. AND, INV) – Update the current (x, y) location of the cell – Update the orientation of the cell

DEF Parser cont..

Class	Type	Methods	Purpose
DEFReader	Rect list<Pin> list<net> list<Component> list<Grid> dict<string,list<Grid> >	readDEF(str) bbox pins nets components gcgrids tracks writeDEF(str)	– Read the .def file specified as a string – Rectangular boundary; – list of boundary pins (Pin objects) – list of nets (Net objects) – list of cells (Component objects) – list of Gcell grids (Grid objects) – dict of tracks (layer:list of Grid objects) – write the updated results into the specified def file